INCH-POUND
MIL-P-24441/38(SH)
23 July 1991

MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, BLACK, FORMULA 160, TYPE IV

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24441.

FORMULA. This formula covers black epoxy-polyamide paint designated Navy Formula 160, type IV, for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

Ingredients	Pounds
Thixotrope 1/	---
Polyamide 2/	30
Polyamide adduct 3/	473
Silica 4/	102
Mineral spirits 5/	110
Butyl alcohol 6/	104

Component B

Ingredients	Pounds
Thixotrope 7/	---
Epoxy resin 8/	709
Silica 4/	62
Black iron oxide 9/	126
Naphtha 10/	147

See footnotes on next page.

AMSC N/A
FSC 8010
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed component tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	11.5	15.5	15.7	19.7	---	---
Volatiles, percent	38.0	42.0	12.0	16.0	---	---
Nonvolatile vehicle, percent	44.5	49.5	65.3	69.3	---	---
Water, percent	---	1.5	---	0.5	---	---
Coarse particles, percent	---	0.3	---	0.3	---	0.5
Consistency, KU	72	83	72	83	72	83
Pounds per gallon	7.9	8.3	10.2	10.6	9.0	9.4
Set to touch, hours						
(a) at 40°F	---	---	---	---	---	3
(b) at 73°F	---	---	---	---	---	3
Dry-hard, hours						
(a) at 40°F	---	---	---	---	---	24
(b) at 73°F	---	---	---	---	---	6
Fineness of grind, NS	3	---	2	---	2	---
Flash point, °F	99	---	100	---	99	---
Titanium dioxide, percent of pigment	20	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	6	---
Sag resistance, mils	---	---	---	---	12	---
Weight per epoxy, vehicle	---	---	175	195	---	---
Contrast ratio, 3 mil dry film	---	---	---	---	0.98	---
Epichlorohydrin content (vehicle), parts per million	---	---	---	2	---	---
VOC, ASTM D 3960, grams per liter 11/	---	---	---	---	---	340

1/ Thixotrope to be used is the manufacturer's choice. In the development of the component A, 10 pounds (1.4 gallons) of Dislon NS-30 made by King Industries were used. Manufacturer is responsible for choosing a thixotrope that meets all the requirements herein, including shelf life. Thixotrope is a pigment for calculation purposes.

2/ GENAMID 2000, General Mills Chemicals; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical.

3/ VERSAMID 280B75, General Mills Chemicals; UNIREZ 2180B73, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical.

4/ Quso WR-55, PQ Corp.

5/ TT-T-291, type III.

6/ Butyl alcohol conforming to ASTM D 304.

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7/ Thixotrope to be used is the manufacturer's choice. In the development of the component B, 10 pounds (1.2 gallons) of Dislon 6500 made by King Industries were used. Manufacturer is responsible for choosing a thixotrope that meets all the requirements herein, including shelf life. Thixotrope is a pigment for calculation purposes.

8/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy Corp.; DER 331, Dow Chemical Corp.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical Corp.; EPI-REZ 510, Celanese.

9/ Bayferrox Black Iron Oxide 318M, Mobay Chemical Corp.

10/ Conforming to ASTM D 3734, type I. In the development of component B, AMSCO Super High Flash Naphtha was used.

11/ For VOC calculations, component A makes approximately 102 gallons and component B makes approximately 102 gallons.

Use of alternate ingredients in this formula must have prior approval of the Naval Sea Systems Command. Approval will be based on review of data demonstrating equivalent physical, chemical and performance characteristics of paint manufactured with the proposed alternate material. Development of the required data package is the responsibility of the proposing agent. Paint incorporating the proposed alternate ingredient shall conform to all the requirements of this military specification sheet and the general specification.

Preparing activity:
Navy - SH
(Project 8010-N257)

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